POWER SERIES

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Name: Solutions

For each of the power series below, find the Radius and Interval of Convergence.

1. $\sum_{n=1}^{\infty} \frac{x^n}{2n-1}$

$$\lim_{n\to\infty} \left| \frac{x^{n+1}}{2(n+1)-1} \frac{2n-1}{x^n} \right| = \lim_{n\to\infty} |x| \frac{2n-1}{2n+1} = |x| < 1 = R \iff -1 < x < 1$$

$\underline{x=-1}$ $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1}$ converges by the Alternating Series Test

$\underline{x=1}$ $\sum_{n=1}^{\infty} \frac{1}{2n-1}$ diverges by Limit Comparison with the Harmonic Series

$$\lim_{n\to\infty} \frac{1}{2n-1}\left(\frac{n}{1}\right) = \frac{1}{2} > 0.$$

Interval of Convergence: $[-1, 1)$

2. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$\lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)!} \frac{n!}{x^n} \right| = \frac{|x|}{n+1} = 0 < 1 \text{ for all } x$$

Interval of Convergence: $(-\infty, \infty)$.

3. $\displaystyle\sum_{n=0}^{\infty} \frac{(x-2)^n}{n^2+1}$

$$\lim_{n\to\infty} \left| \frac{(x-2)^{n+1}}{(n+1)^2+1} \frac{n^2+1}{(x-2)^n} \right| = \lim_{n\to\infty} |x-2| \frac{n^2+1}{n^2+2n+2} = |x-2| < 1 = R \iff -1 < x-2 < 1 \iff 1 < x < 3.$$

$\underline{n=1}$

$\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$ converges by the Alternating Series Test.

$\underline{n=3}$

$\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^2+1}$ converges by Limit Comparison With $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$: $\displaystyle\lim_{n\to\infty} \frac{1}{n^2+1} \frac{n^2}{1} = 1 > 0.$

Interval of Convergence: $[1,3]$.

4. $\displaystyle\sum_{n=2}^{\infty} \frac{(x+2)^n}{2^n \ln(n)}$

$$\lim_{n\to\infty} \left| \frac{(x+2)^{n+1}}{2^{n+1}\ln(n+1)} \frac{2^n \ln(n)}{(x+2)^n} \right| = \lim_{n\to\infty} |x+2| \left(\frac{1}{2}\right) \frac{\ln(n)}{\ln(n+1)} \overset{L'H}{=} \lim_{n\to\infty} \frac{|x+2|}{2} \frac{1/n}{1/n+1} = \lim_{n\to\infty} \frac{|x+2|}{2} \frac{n+1}{n} = \frac{|x+2|}{2}$$

$\frac{|x+2|}{2} < 1 \iff |x+2| < 2 \iff -2 < x+2 < 2 \iff -4 < x < 0.$

$\underline{x=-4}$ $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n 2^n}{2^n \ln(n)} = \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln(n)}$ converges by the Alternating Series Test.

$\underline{x=0}$ $\displaystyle\sum_{n=2}^{\infty} \frac{2^n}{2^n \ln(n)} = \sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ Diverges by comparison with the Harmonic Series!

$\frac{1}{n} \le \frac{1}{\ln(n)} \iff \ln(n) \le n \iff \ln(n) - n \le 0$

$f(x) = \ln(x) - x \Rightarrow f'(x) = \frac{1}{x} - 1 < 0$ for all $x > 1 \Rightarrow f(x)$ decreasing on $(1,\infty)$.

Since $f(e) = \ln(e) - e = 1 - e < 0$, it follows that for all $n > 2$

$$f(n) = \ln(n) - n < 0.$$

Interval of Convergence: $[-4,0)$.